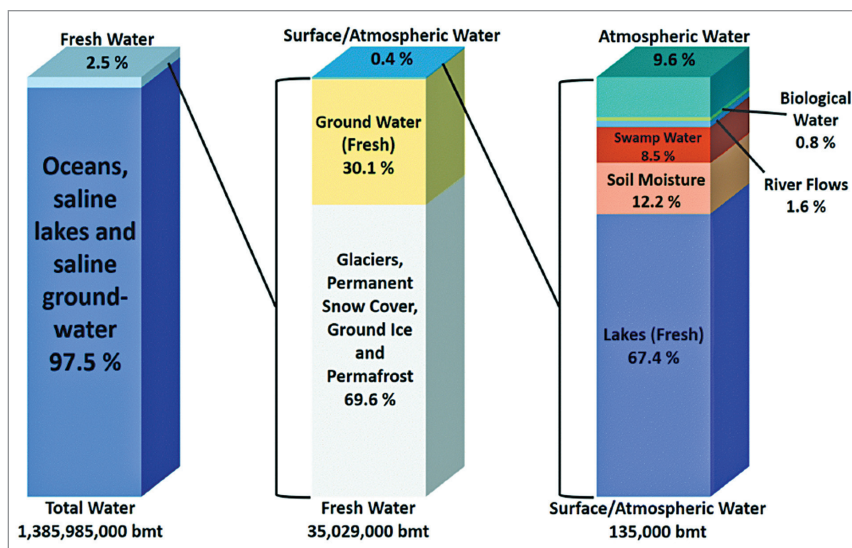


While we confront a dwindling water supply that has transitioned from a looming threat to a stark reality, a beacon of hope emerges: the innovative technology of extracting water from the air. This advancement promises to alleviate the escalating water crisis and redefine our relationship with this essential resource. Together, we are witnessing the rise of a promising field that can address water shortages, provide clean drinking water in remote or arid areas, and lessen the environmental burden of conventional water sources. Let us explore the significance of this technology, its benefits, challenges, and crucial role in ensuring a water-secure future for all.

The principle behind water-from-air technology is like that of a dehumidifier, but it is far more advanced and efficient. From compact units for home use to large-scale systems, this technology is versatile, catering to communities of all sizes. Imagine a world where water availability is not constrained by location or weather conditions, where every community can thrive with a consistent supply of clean water drawn from the air. This vision is becoming a reality, from the vibrant streets of Bengaluru to tech hubs worldwide, where both Indian and international companies are pioneering the transformation of this once science-fiction concept into a practical solution.

The science behind 100% renewable water-from-air technology

To understand the innovative nature of 100% renewable water-from-air technology, it is essential to consider what makes this water source truly renewable. Swapnil Shrivastav, Co-Founder & CEO of Uravu Labs Pvt Ltd, explains, "This concept hinges on two pivotal elements. Firstly, the



The abundance of atmospheric water (Credit: Peeters, R. et al. Environ. Sci. Water Res. Technol 2020, 6, 2016-2034)

Earth's atmosphere contains approximately six times the volume of water found in all the world's rivers combined, making it an boundless resource. Secondly, the energy needed to operate these devices is entirely derived from renewable sources such as solar power, waste heat, hydroelectricity, and more. This combination of abundant atmospheric moisture and sustainable energy sources ensures that the water extracted remains 100% renewable."

With technological advancements, various innovative solutions have emerged to address water scarcity by extracting moisture from the atmosphere. These diverse approaches leverage scientific principles and engineering ingenuity, providing multiple avenues to combat water shortages.

One method is refrigeration-based atmospheric water generators (AWGs), which use a refrigeration cycle to cool the air, causing water vapour to condense into a liquid state. This liquid is then meticulously filtered and subjected to UV sterilisation to ensure its safety for consumption.

In contrast, desiccant-based

AWGs use unique materials called desiccants to absorb moisture from the air. Once absorbed, the moisture is extracted from the desiccant and purified to meet potable water standards.

Membrane-based AWGs take a slightly different approach, employing specialised materials that selectively allow water vapour to pass through while separating it from other atmospheric gases. The isolated water vapour is subsequently purified to ensure its suitability for drinking.

Each of these technologies offers a unique and practical approach to harnessing water from the atmosphere, contributing to sustainable water resources in a world grappling with water scarcity challenges.

As Shrivastav mentions, "They employ a technology grounded in the principles of desiccants, materials well-known for their moisture-absorbing capabilities." Desiccants like silica gel, commonly found in packaging, function similarly. Uravu Labs uses calcium chloride, Earth's second most abundant salt, as the desiccant. This material draws moisture from the air, but the main